Problem F. False Amber

Input file:	`standard input`
Output file:	`standard output`
Time limit:	2 seconds
Memory limit:	1024 megabytes

Ras and *Salt* are playing a simplified version of Texas Hold'em called CAFE MiLK Poker. The game uses $2n$ cards with ranks $1 \sim 2n$, and proceeds as follows:

- One chip is placed in the center of the table. Each player has m chips and n cards.
- *Ras* and *Salt* each draw one card **uniformly at random** from their own deck.
- Both players know the two complete decks and the card they drew, but neither knows the card drawn by the other player.
- *Ras* may choose to **check** or go **all-in**.
- If *Ras* checks, both players reveal their cards, and the player with the higher card takes the chip on the table.
- If *Ras* goes all-in:
  - *Salt* may choose to **fold** or **call**.
  - If *Salt* folds, *Ras* takes the chip on the table.
  - If *Salt* calls, both players reveal their cards, and the player with the higher card takes a total of $m + 1$ chips from the table and the other player.

Ras and *Salt* may each use a **mixed strategy** $\pi = p_i$: whenever the player makes a decision while holding card i, they choose one action with probability p_i and the other action with probability $(1 - p_i)$. Both players want to maximize their own **expected profit**, defined as the number of chips they have at the end of the game minus m.

You are *Chiffon*, *Ras*'s best friend, and you know that *Salt* can read minds: although she cannot know which card *Ras* drew, she can know *Ras*'s strategy! Therefore, you need to help *Ras* find a strategy that maximizes her expected profit. Formally, you need to choose a mixed strategy π_r for *Ras* that maximizes *Ras*'s expected profit when *Salt* may choose her strategy π_s based on π_r.

Input

This problem contains multiple test cases.

The first line contains a positive integer T ($1 \le T \le 10^4$), representing the number of test cases.

For each test case:

- The first line contains two integers n, m ($1 \le n \le 2 \times 10^5$, $1 \le m \le 10^9$).
- The second line contains n integers a_i ($1 \le a_i \le 2n$), representing *Ras*'s cards.
- The third line contains n integers b_i ($1 \le b_i \le 2n$), representing *Salt*'s cards.

It is guaranteed that the sum of n over all test cases does not exceed 2×10^5, and that all a_i and b_i are pairwise distinct; in other words, every integer from 1 through $2n$ occurs exactly once.

Output

For each test case, output one line containing a floating-point number representing the answer.

Your answer is considered correct if its absolute or relative error does not exceed 10^{-6}. Formally, let your answer be a and the correct answer be b. Your answer is accepted if and only if $\frac{|a-b|}{\max(1,|b|)} \le 10^{-6}$.

Example

standard input	standard output
3 1 1 1 2 2 2 1 4 2 3 3 10 3 6 1 5 2 4	0.000000000000000 0.833333333333333 0.747474747474747

Note

In the first test case, *Ras* never goes all-in.

In the second test case, *Ras* goes all-in with probability $\frac{2}{3}$ when she draws card 1, and always goes all-in when she draws card 4. One possible strategy for *Salt* in this case is to always call. Therefore:

- With probability $\frac{1}{6}$, *Ras* draws card 1 and checks, earning a profit of 0.
- With probability $\frac{1}{3}$, *Ras* draws card 1 and goes all-in, earning a profit of -2.
- With probability $\frac{1}{2}$, *Ras* draws card 4 and goes all-in, earning a profit of 3.

Her expected profit is $\frac{1}{6} \cdot 0 + \frac{1}{3} \cdot (-2) + \frac{1}{2} \cdot 3 = \frac{5}{6}$.